التمريض
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جامعة حلوان

Introduction to Evidence Based Practice in Nursing

Out lines:

- Introduction.
- Definitions of key terms.
- Benefits of EBP.
- Barriers to EBP in nursing.
- Process of EBP.
- Application of Evidence Based practice.
- Levels of Evidence Based practice.
- Common sources of Evidence Based practice.
- Strategies to promote evidence-based practice / evidence-informed decision-making b nurses.
- **References.**

Goal:

To empower students knowledge, practice and positive attitude towards Evidence Based practice.

Objectives:

At the end of this lecture, the participants will be able to:

- Define each key term.
- Discuss benefits of Evidence Based practice.
- Identify barriers of Evidence Based practice.
- Enumerate steps of Evidence Based practice.
- Apply Evidence-based practice in nursing.
- Identify levels of Evidence Based practice.
- Have a positive attitude toward evidence based practice.

Introduction

Nurses use different sources of knowledge to make clinical decisions. Intuition, trial and error, tradition, authority and clinical experience are often used as sources of knowledge are reliable and all don't produced desired outcomes (Ledbetter and Stevens, 2000).

A procedure done by trial and error might be performed successfully sometimes and other times might not. Intuition and authority may lead to faulty clinical decision making. Although experience can be a good teacher, it can contain bias. For example, just because the nurse has experience in successfully performing an intervention a certain way doesn't mean it is the best way or that will be successful every time.

Nowadays there is comprehensive databases are available through different internet sites to assist nurses in applying the most recent best evidence to their clinical practice . Research evidence provides scientific basis for practice.

What the Evidence-based Practice (EBP)

- Evidence-based practice is an approach that promotes the collection, interpretation, and integration of valid, important, and applicable patient-reported, clinician-observed, and research-derived evidence. In addition to the collection, evaluation, and interpretation of data, EBP emphasizes the dissemination of information so that the evidence can reach clinical practice. *(McKibbon, et al., 1995).*

Definition of evidence based practice

- EBP is a systematic approach for problem solving for health care providers, characterized by the use of the best currently available for clinical decision – making in order to provide consistent and best possible care to the patient.

- EBP is a care that integrates best scientific evidence with clinical expertise, knowledge and decision making preferences of patients.

Evidence-based medicine

- Is the conscientious, explicit and judicious used of current best evidence in making decisions about the care of individual patients.

Evidence based nursing

- Evidence based nursing is a process by which nurses make clinical decisions using the best available research evidence, clinical expertise and patient preferences.

Benefits of EBP

1) The objective of (EBP) is to apply the best available evidence in clinical care.
2) Improve the quality of patient care.
3) Increase staff member skills and confidence.
4) Improve management and quality of educational programs.
5) Reduce cost and improve outcomes.
6) Help staff to apply new technologies.
7) Keep practice current and relevant.
8) Policies and procedures are current and include the latest research.

Barriers to EBP in nursing

1) Research related barriers

2) Nurse-related barriers

3) Organizational barriers

1) Research related barriers:

- Research is seen as too complicated, too scholarly, excessively statistical, ambiguous, and having limited or no relevance to practice.

- Research reports lack clear practice implications and generalizability.

2) Barriers related to nurse characteristics:

- Lack of the knowledge needed to interpret statistical analyses.
- Lack of interest.
- Lack of confidence in critical appraisal skills.
- Lack of knowledge and skills to confidently conduct computer based literature searches and utilize the research process.
- Laziness/lack of motivation/ burnout
- Nurse's perceptions that they lack the authority and cooperation to change patient care procedures.
- Negative beliefs, attitudes and values

Barriers related to organizational characteristics:

- Limited or lack of time.
- Heavy patient workloads.
- Inadequate staffing.
- Limited access to resources.
- Lack of support from nurse managers.
- Different goals for practice between administrators and staff nurses.
- Lack of evidence-based practice mentors in health-care systems.

The process of EBP (5A)

1. **Ask:** Converting information needs into clear questions
2. **Acquire:** Seeking evidence to answer those questions
3. **Appraise:** Evaluate the evidence for its validity and usefulness.

4. **Apply:** Integrating findings with clinical expertise, patient needs, applying the finding.

5. **Assess**: Evaluating performance.

Step (1) Ask

Formulating questions (PICOT):

- **P**: Problem or patient
- **I**: Intervention
- **C:** Comparison Intervention
- **O:** Outcomes
- **T:** Time

Step (2) Acquire

- Selecting resources and research for the evidence.

Step (3) Appraise

- In this step the nurse critically appraise the evidence for validity, impact, and usefulness in clinical practice.
- Several issues should be considered, particularly the transferability of the innovation, the feasibility of implementing it, and the cost-benefit ratio

Step (4) Apply

- Application of the evidence immediately follows the critical evaluation. The main question that arises at this stage is how the evidence can be applied to meet specific situation or need.

Step (5) Assess

- After applying what the nurse learned from the literature to the situation, nurse will need to evaluate.
- Nurse notice changes of patient care and outcomes.

- The nurse should not only assess the outcomes but also the effectiveness and the efficiency of the evidence based process.

Levels of Evidence:

- Systematic Reviews
- Meta-Analyses
- Well-designed Experimental Studies (RCTs)
- Well-designed Nonexperimental Studies
- Observational Descriptive Studies
- Expert Consensus
- Published Case Reports and Clinical Examples

Common sources of evidence:

- Patient values and preferences : Inclusion of the patient in decisions about his or her health care .
- Textbooks – provide a knowledge base.
- **Strategies to promote evidence-based practice / evidence-informed decision-making by nurses:**
  - Provide nurses with access to a rich library with nursing journals.
  - Provide nurses with opportunities for working with a computer and for searching the Internet in the workplace.
  - System support for searching and reading professional literature.
  - Implementation of a virtual journal club.
  - Provide nurses with access to evidence-based practice resources via mobile information technologies.
  - Implement an EBP mentorship programme with EBP mentors who are "skilled in both EBP and organizational culture and change.
  - Support nurses to acquire the skills needed to read, evaluate and critically

appraise evidence.

- Establish leadership, a coherent change strategy, and relationships between point of care providers and managers.
- Nurse Managers act as role models.
- Nurse Managers provide the resources and the support for the work and celebrate success with recognition of unit staff.
- Designate a champion who is accessible to the nurses, along with other leaders and innovators who can answer questions and reinforce the practice change.
- Involve the “clinical educator as a part of the support system of the EBP change.